Linear Algebra - 2

Recap :-

→ Dataset (X, Y) → X → Features (Variables) / Independent Variables (input)

y → Label (Target) / Dependent Variable

→ Task → classifier (classification) ✓

→ mapping function $[y = f(x)]$

→ $w_1 x_1 + w_2 x_2 + w_0 = 0$.

$[w_0, w_1, w_2]$ → weights / parameters.

Agenda :-

[Linear Algebra ✓ Matrices] + ["vectors" ✓]

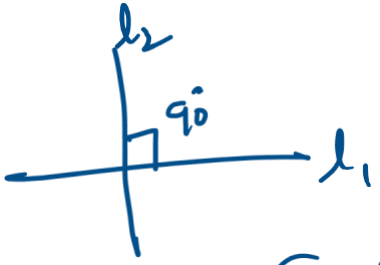
$f(x) = [w_0 + w_1 x_1 + w_2 x_2 + \cdots w_n x_n]$?

→ w [ ] [x] ?

$y = w_1 x_1 + w_2 x_2 + w_0$ $\quad w_1 x_1 + w_2 x_2 + w_0 = 0$ $\quad y = mx + c$

→ m_1 m_2

l_1 and l_2



$$m_1 \times m_2 = -1$$

$$\frac{3}{2} \times \frac{-2}{3} = -1$$

$y = mx + c$ (3/2)

Vectors

Vector → Magnitude

Direction

(Data point)

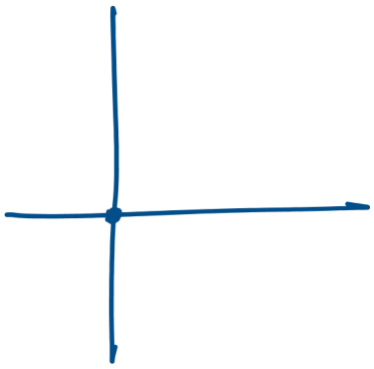


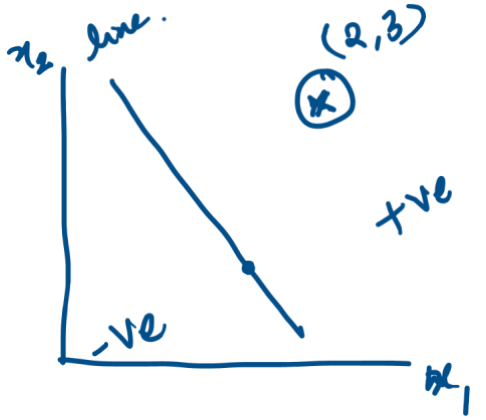
latitudes

longitudes.

$\hat{y} = w_1 x_1 + w_2 x_2 + w_0 = 0$

$y = f(x)$





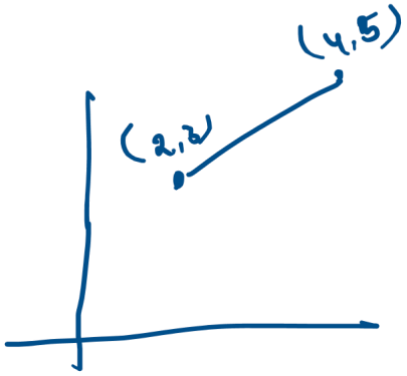
$$w_0 + w_1 x_1 + w_2 x_2 > 0$$

$$y = w_0 + w_1(x_1) + w_2(x_2)$$

$(y) = f(x)$

y - output, $\{x_1, x_2, x_3 - - x_n\}$ Inputs

Vectors





latitudes

longitudes

$$\sqrt{(2-0)^2 + (3-0)^2}$$

Euclidean distance.

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

+ magnitude of vector

origin → data point

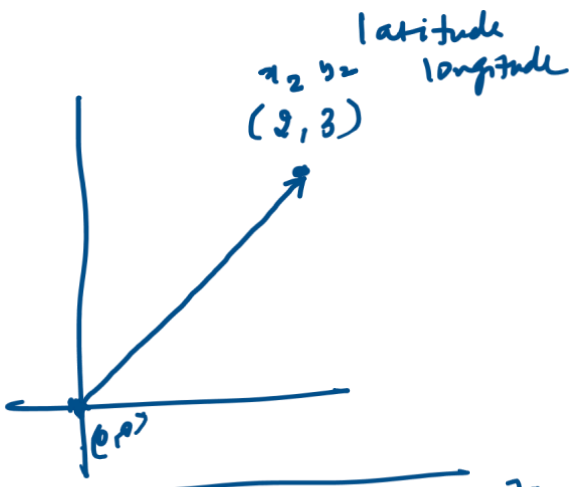
mag $\sqrt{x^2 + y^2}$

$\vec{x}$, $\bar{x}$

$\underrightarrow{x}$ → Magnitude $\|x\| = \sqrt{x_1^2 + x_2^2}$.

L_2 - Norm - Euclidean distance

Vector → data point



✓ Magnitude of a vector.

Vector → direction
Magnitude.

Euclidean dist $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\sqrt{x^2 + y^2}$

$\vec{x}$ (2,3) → $\sqrt{2^2 + 3^2}$

$||\vec{x}||_2$ → L_2 Norm - Euclidean distance from origin to the datapoint.

$\vec{x}$ (2, 3, 4) → $\sqrt{2^2 + 3^2 + 4^2}$

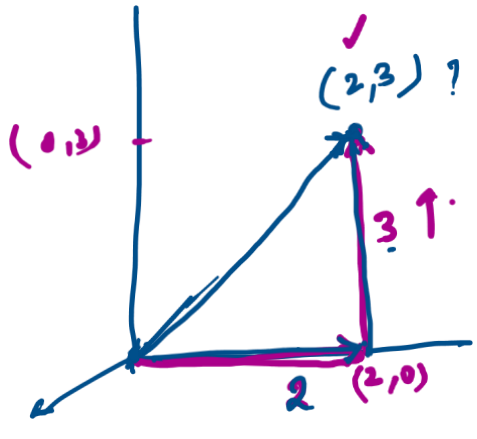
$\vec{x}$ (2, 3, 4, ...) → $\sqrt{2^2 + 3^2 + 4^2 \cdots q^2}$ (d)

L_2 Norm = Magnitude = displacement = $\sqrt{x_1^2 + x_2^2 + \cdots x_d^2}$

L_1 Norm

dista = $|2| + |3|$

$|x_1| + |x_2|$



L_1 Norm → (of distances) - Manhattan distance

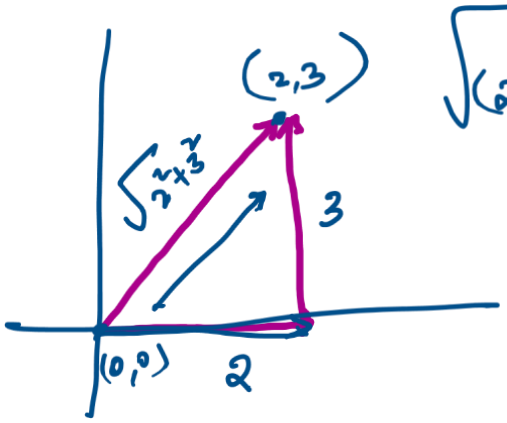
L_1 Norm → $|x_1| + |x_2| + \cdots |x_d|$.

L_2 Norm $\|\vec{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots x_d^2}$

L_1 Norm $|\vec{x}|_2 \rightarrow |x_1| + |x_2| + |x_3| + \cdots |x_d|$

$\|\vec{x}\|_2 = \sqrt{2^2+3^2}$

$|\vec{x}|_1 = |2| + |3| \rightarrow 5$



$\sqrt{(2-0)^2 + (3-0)^2}$

$[2, 3]$

Vector $\vec{x} \rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ Column

$\vec{x} \rightarrow \begin{bmatrix} x_1 \\ x_n \end{bmatrix}$ → Column

Matrices

$[\,\,]$ $[\,\,]$

2×3 2×3

$2 \times 3 \quad 3 \times 2 \rightarrow [\,\,]_{2 \times 2}$

$[\,\,]$ $[\,\,]$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times2} \quad \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2\times2} \quad \begin{bmatrix} 1\times5+2\times7 & 1\times6+2\times8 \\ 3\times5+4\times7 & 3\times6+4\times8 \end{bmatrix}_{2\times2}$$

$m \times n \quad n \times p$ → Resultant → $m \times p$ ✓

Data set → observations – Data point

$[X]$ – Data points →

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ – Vector

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_{2\times1} \quad \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{2\times1}$

$\begin{bmatrix} 2 & 1 \end{bmatrix}_{1\times2} \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{2\times1}$

→ $w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 - \cdots$

Mapping function.

$w_1 \quad w_2 \quad (w_0)$

$\begin{bmatrix} 1 & 1 & -98 \end{bmatrix}$ → Weights / parameters

$\begin{bmatrix} 33 & 27 \end{bmatrix}$

$\begin{bmatrix} 13 & 6.5 \end{bmatrix}$

$\begin{bmatrix} 47 & 39 \end{bmatrix}$

⋮

w_1 w_2 x_1 x_2

$\begin{bmatrix} 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 33 & 27 \\ 13 & 6.5 \\ 43 & 39 \end{bmatrix}$

rows

$\begin{bmatrix} w_1 & w_2 \\ 1 & 1 \end{bmatrix}_{1\times 2} \quad \begin{bmatrix} 33 & 27 \end{bmatrix}_{1\times 2}$

Cols.

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2\times 1} \quad \begin{bmatrix} 33 \\ 27 \end{bmatrix}_{2\times 1}$

Column Vectors

$\vec{w} \quad \vec{x}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 33 \\ 27 \end{bmatrix} \rightarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\vec{w} \quad \vec{x}$

$\vec{w}^T \cdot \vec{x} \qquad \left(\begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$

$w_1 x_1 + w_2 x_2 + w_0$?

$\vec{w} \begin{bmatrix} w_1 & w_2 & w_3 & \cdots & w_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$

$w^T \cdot \vec{x} + w_0$

$\vec{w} \begin{bmatrix} w_0 & w_1 & w_2 & w_3 & \cdots & w_m \end{bmatrix} \begin{bmatrix} x_0 = 1 \\ x_1 \\ x_2 \\ x_3 \\ \vdots \end{bmatrix}$

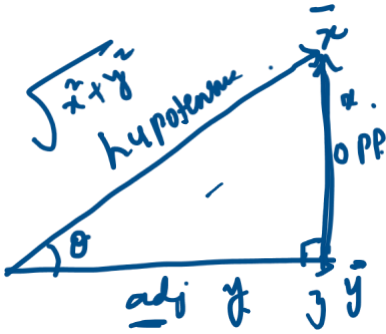
$$\left[w_0 x_1 + w_1 x w_1 + w_2 x_2 + \cdots w_n x_n \right]$$

→ W is a collection values → $\vec{W}$

$\vec{x}$ $\vec{y}$ $\left[\vec{x}^T \cdot \vec{y}\right]$

Dot product - $\vec{x}^T \cdot y$?





$$\cos\theta = \frac{x^T \cdot y}{\|x\| \, \|y\|}$$

$$\cos\theta = \frac{adj}{hyp} \cdot \frac{\vec{y}}{\sqrt{x_1^2 + y^2}} \frac{\|y\|}{}$$

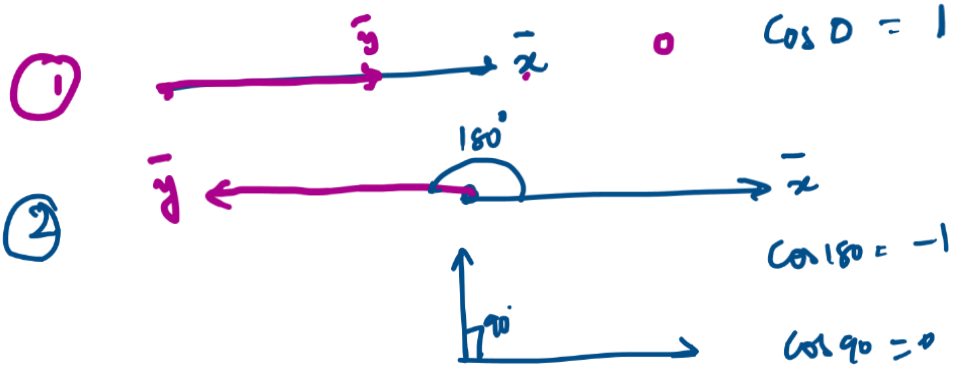
$$\cos\theta = \frac{adj}{hyp}$$

$$\sin\theta = \frac{opp}{hyp}$$

$$\tan\theta = \frac{opp}{adj}$$

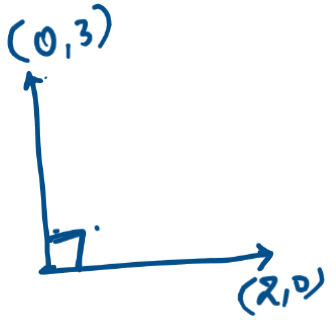
→ Vector Algebra.

Similarity of vectors. (Cosine Similarity)



$$\cos\theta = \frac{\vec{x}^T \cdot \vec{y}}{\|x\| \, \|y\|} \rightarrow$$ direction of two vectors $\vec{x}, \vec{y}$

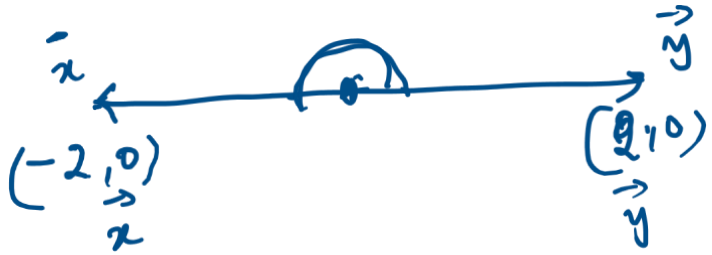
$\vec{x}^T \cdot \vec{y}$ — dot product.



$$\vec{x}^T \quad \begin{bmatrix} 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\cos 90^\circ = \frac{0 \times 2 + 3 \cdot 0}{\|x\| \, \|y\|} = \frac{0}{\|x\| \, \|y\|}$$

$$\cos\theta = \frac{\vec{x}^T \cdot \vec{y}}{\|x\| \, \|y\|}$$



✓

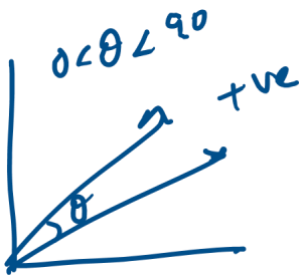
$$\cos\theta = \frac{\vec{x}^T \cdot \vec{y}}{\|x\| \, \|y\|}$$ — derivations

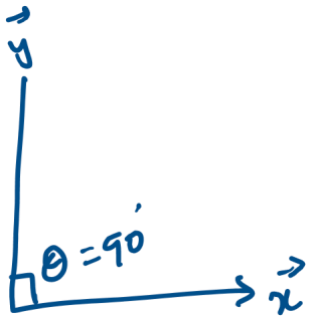
$$\vec{x}^T \quad \begin{bmatrix} -2 & 0 \end{bmatrix} \qquad \vec{y} \quad \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\frac{-4+0}{2+2} = \frac{-4}{4} = -1 \checkmark$$

$\cos 180$ ✓

→ Alignment of vectors ✓





dot product = 0

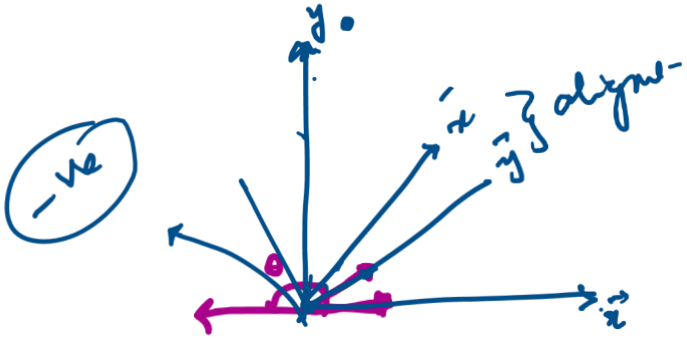
A B

[ ] [ ]θ

1×2 2×1

⇒ A[0]B[0] + A[1] B[1] ✓

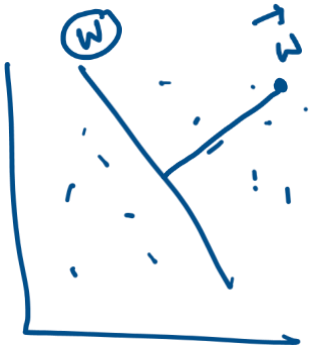
Dot Product :- Alignment of vector with another vector.



dot product

$W^T \cdot \vec{x}$

$W^T \cdot \vec{x} + w_0$



Unit Vector → Norm = 1

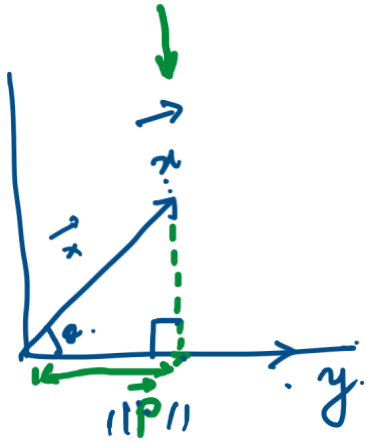
Vector magnitude $||\vec{x}|| = 1$ unit vector

Vector → unit vector $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$

$[w_1 \; w_2]$

→ unit $\begin{bmatrix} \frac{w_1}{\sqrt{w_1^2+w_2^2}} \\ \frac{w_2}{\sqrt{w_1^2+w_2^2}} \end{bmatrix}$ → unit Vector

→ Projection of a vector.



angle $\vec{x}$ b/w $\vec{x}$ and $\vec{y}$ =

angle between $\vec{x}$ and $\vec{P}$

$$\cos\theta = \frac{\vec{P}}{||\vec{x}||} \Bigg| = \cos\theta = \frac{\vec{x}^T \cdot \vec{y}}{||x|| \cdot ||y||}$$

$$\frac{\|\vec{P}\|}{\cancel{A}\cancel{x}\cancel{T}} = \frac{\vec{x}^T \cdot \vec{y}}{\cancel{\|x\|} \cdot \|y\|} \qquad \|\vec{P}\| = \frac{\vec{x}^T \cdot \vec{y}}{\|y\|}$$

Projection of a vector $\vec{x}$ onto another vector $\vec{y}$

$$\|\vec{P}\| = \frac{\vec{x}^T \cdot \vec{y}}{\|y\|} \quad \checkmark$$

$$\cos\theta = \frac{\vec{x}^T \cdot \vec{y}}{\|x\| \, \|y\|}$$ - alignment of vectors $\vec{x}$ and $\vec{y}$

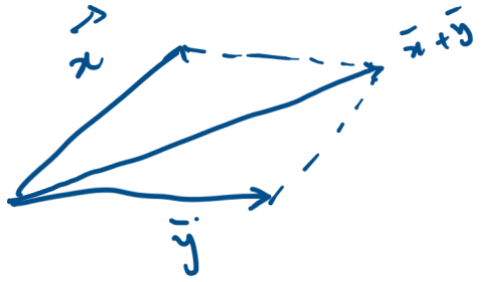
Projection of a vector $\vec{x}$ onto another vector $\vec{y}$

$$\|P\| = \frac{\vec{x}^T \cdot \vec{y}}{\|y\|} \quad - (2)$$

Additional Notes

Vectors Basics.

① addition. $\vec{x}$, $\vec{y}$ Parallelogram rule



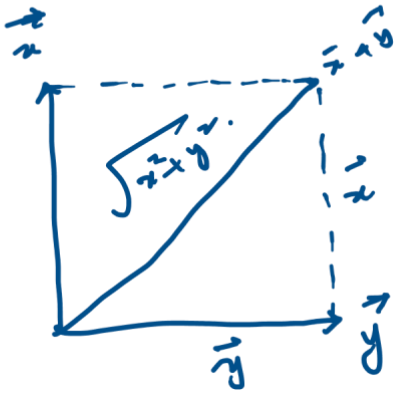
$$\vec{z} = \vec{x} + \vec{y}$$

The Magnitude of the resultant vector $\vec{z}$

$$= \sqrt{|x|^2 + |y|^2 + 2|x||y|\cos\theta}$$ — law of cosines

② if vectors are perpendicular $\vec{x} \perp \vec{y}$

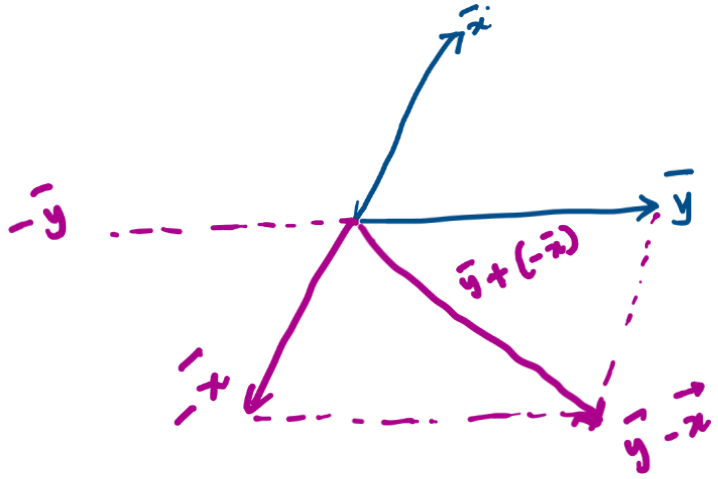
Magnitude of $(\vec{x}+\vec{y}) = \sqrt{|x|^2+|y|^2}$



$\cos\theta = 0$

$\theta = 90^\circ$

→ rules for Substraction of vectors $\vec{x}$ and $\vec{y}$



$$X - Y = X + (-Y)$$

Magnitude of the resulting vector $\vec{x} - \vec{y}$

$$= \sqrt{|x|^2 + (y)^2 - 2|x||y| \cos\theta}$$

Proof for $\cos\theta = \dfrac{x^T \cdot y}{\|x\| \cdot \|y\|}$

Consider two vectors $\vec{x}$ and $\vec{y}$

Look at $X - Y$ (Substraction rule)

Now take Squared magnitude $\|X - Y\|^2$

This is nothing but the magnitude of the $X - Y$ vector in Squared form.

Alternate way (using dot products)

$$\|x-y\|^2 = (x-y)\cdot(x-y)$$

$$= x\cdot x - 2x\cdot y + y\cdot y$$

$$= \|x\|^2 + \|y\|^2 - 2x\cdot y \quad -(1)$$

from Geometry (law of cosines)

$$\|(x-y)\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta \quad -(2)$$

Equation (1) and (2) must be Same

$$\cancel{\|x\|^2} + \cancel{\|y\|^2} - 2\cdot x\cdot y = \cancel{\|x\|^2} + \cancel{\|y\|^2} - 2\|x\|\|y\|\cos\theta$$

$$x\cdot y = \|x\|\,\|y\|\cdot\cos\theta$$

$$\boxed{\frac{x\cdot y}{\|x\|\,\|y\|} = \cos\theta}$$

Note:- Read law of cosines from trigonometry if required.